

1. Use separation of variables to solve the differential equation

$$\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$$

with boundary condition $u(0, y) = 8e^{-3y} + 4e^{-5y}$.

2. Solve the following boundary value problems by separation of variables:

(a) $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$ with $u(x, 0) = 4e^{-x}$

(b) $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial y} + u$ with $u(x, 0) = 3e^{-5x} + 2e^{-3x}$

3. Solve the following heat equation problem by separation of variables:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad \text{with} \quad \frac{\partial u(0, t)}{\partial x} = 0, \quad u(2, t) = 0, \quad u(x, 0) = 8 \cos\left(\frac{3\pi x}{4}\right) - 6 \cos\left(\frac{9\pi x}{4}\right)$$

4. Solve the following wave equation problem by separation of variables:

$$\frac{\partial^2 y}{\partial t^2} = 4 \frac{\partial^2 y}{\partial x^2} \quad \text{with} \quad y(0, t) = y(5, t) = 0, \quad y(x, 0) = 0, \quad \frac{\partial y(x, 0)}{\partial t} = 5 \sin(\pi x)$$